

Basics of Sacral Diagnosis

Landmarks and Landmark Symmetry,
Static Exam and Special Tests

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Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Session Objectives

1. Be able describe the basic sacral landmarks and their locations
2. Be able to describe how to perform and interpret the different motion tests for sacral diagnosis.
3. Be able to describe the different sacral axes and their significances.
4. Be able to describe physiologic and non-physiologic sacral dysfunctions and what types of sacral dysfunction fall into each category and how they are named
5. Be able to apply the rules of L5 on the sacrum to make a sacral diagnosis if only given an L5 diagnosis.

Sacrum 101 Outline

1. Context
2. Anatomy and Landmarks
3. Axes of Motion
4. Osteopathic Examination
5. Special Tests
6. List SD Diagnosis Categories

Sacrum: Clinical Significance

Pediatrics/Internal Medicine

- GI
- GU
- Pulmonology

OB/GYN and Urology

- Ante-partum, labor
- Dysmenorrhea
- Fibroids
- Fertility
- Erectile Dysfunction
- Urinary Tract Infections

Surgery/Trauma

- Injuries
- Pre/Post Op: Parasympathetic Nervous System & lymph stasis

Musculoskeletal

- Back pain
- Cephalgia
- Leg length Discrepancy
- Gait

Psych

- Sacral sag- fascial drag

Embryology/ Development

- Five Segments
- 35 centers of ossifications
- Potential for Intraosseous strain
- Most segments fuse by age 25 to 35-years-old
- S1-S2 may not fuse until 4th decade or later

Architectural Considerations

- Suspended between the ilia
- Connected to the ischia
- Connected to the coccyx
- Supporting Spinal Column
- Pelvic Structure resting on B/L Lower Extremity via femoral heads

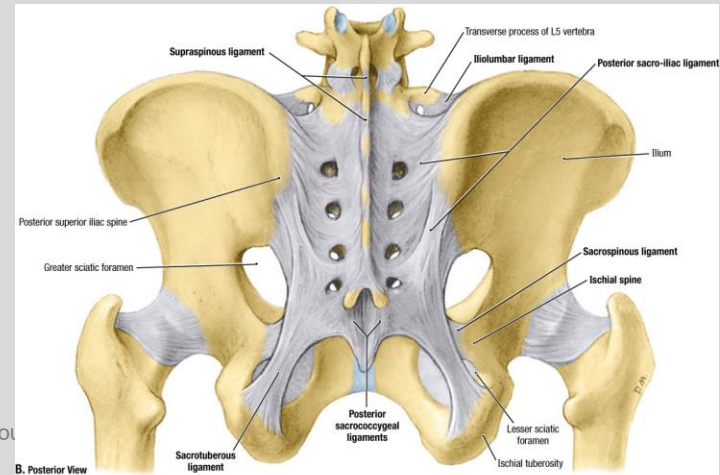
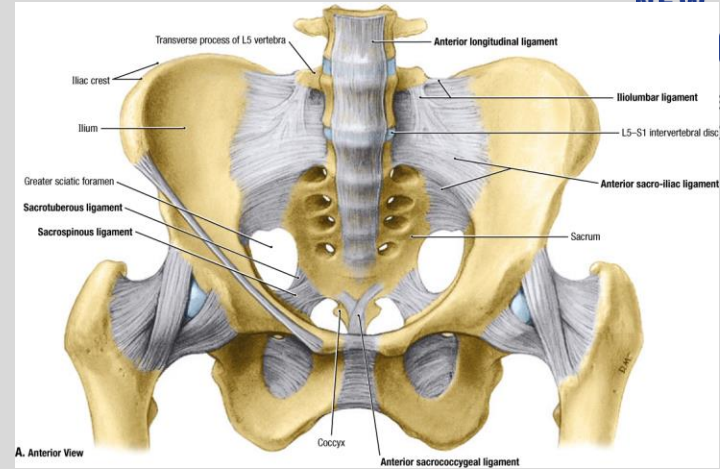
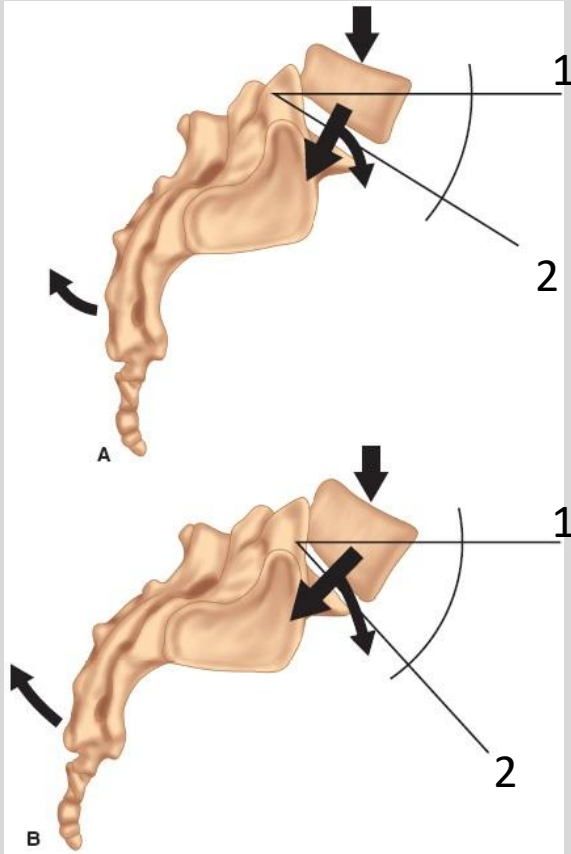


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Lumbosacral Junction



Lumbosacral Angle (Ferguson's Angle)

between:

1-horizontal

2-sacral base

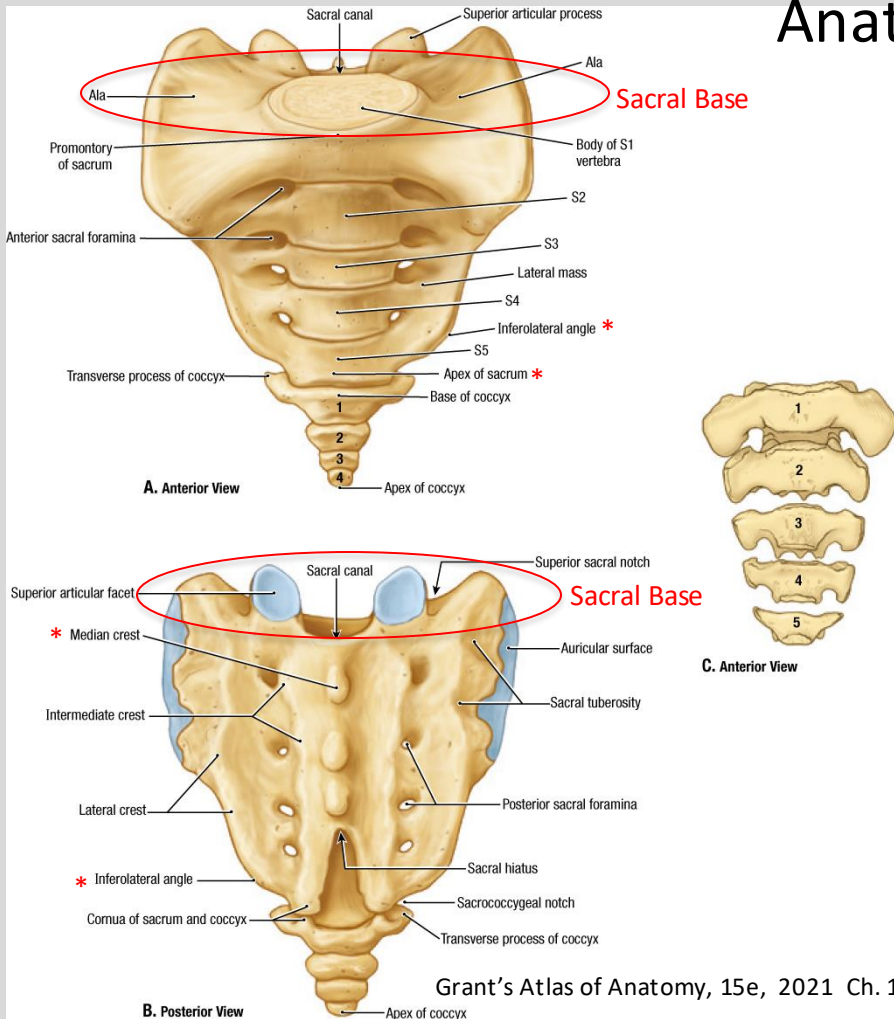
Normal= Approximately 25° - 35°

An increase in the lumbosacral angle creates a greater anterior vector force, which increases the lumbosacral strain.

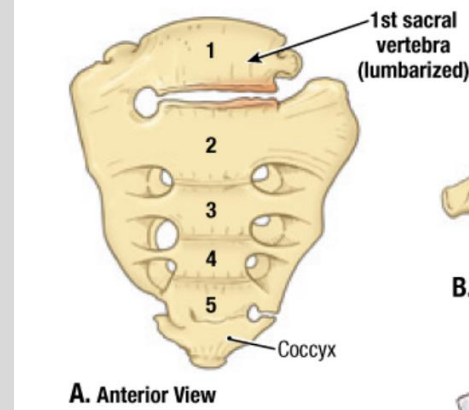
Anatomy

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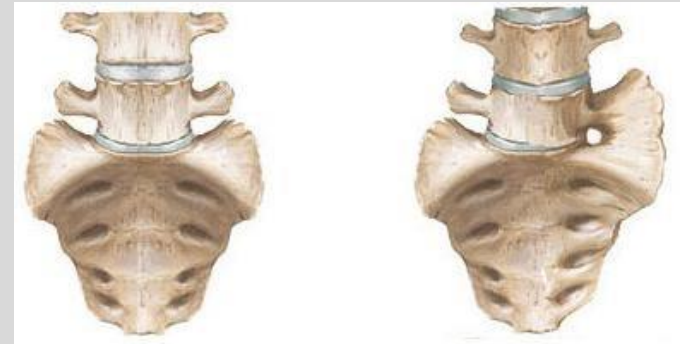


Lumbarization of S1



Grant's Atlas of Anatomy, 15e, 2021 Ch. 1

Sacralization of L5

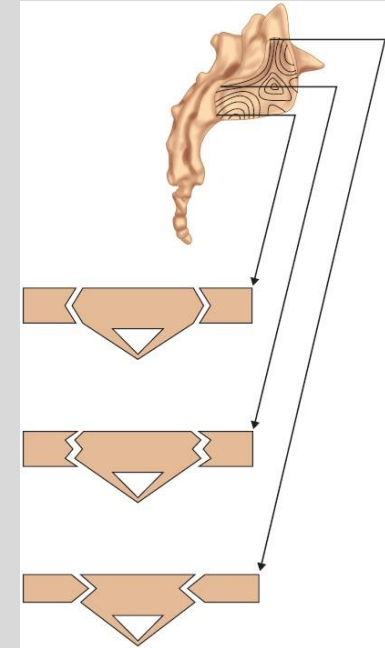
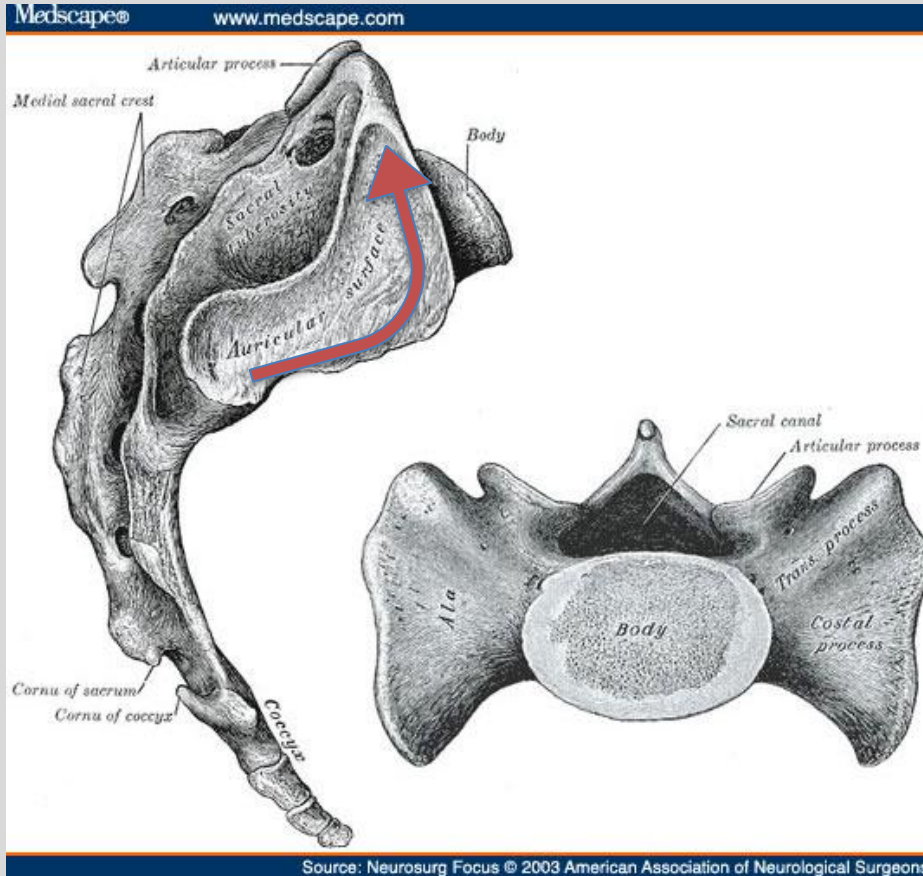


<https://mobilephysiotherapyclinic.in/sacralization-and-physiotherapy-management/>

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Sacraliliac joint

Sacroiliac articulation configurations
as seen at different sacral levels



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Somatic Dysfunction of the Sacrum may have an influence on Nerve structure and function

- Ventral nerve roots
- Parasympathetics - S2-4
- Sympathetic Chain Ganglia
 - Ganglion Impar @ sacrococcygeal junction

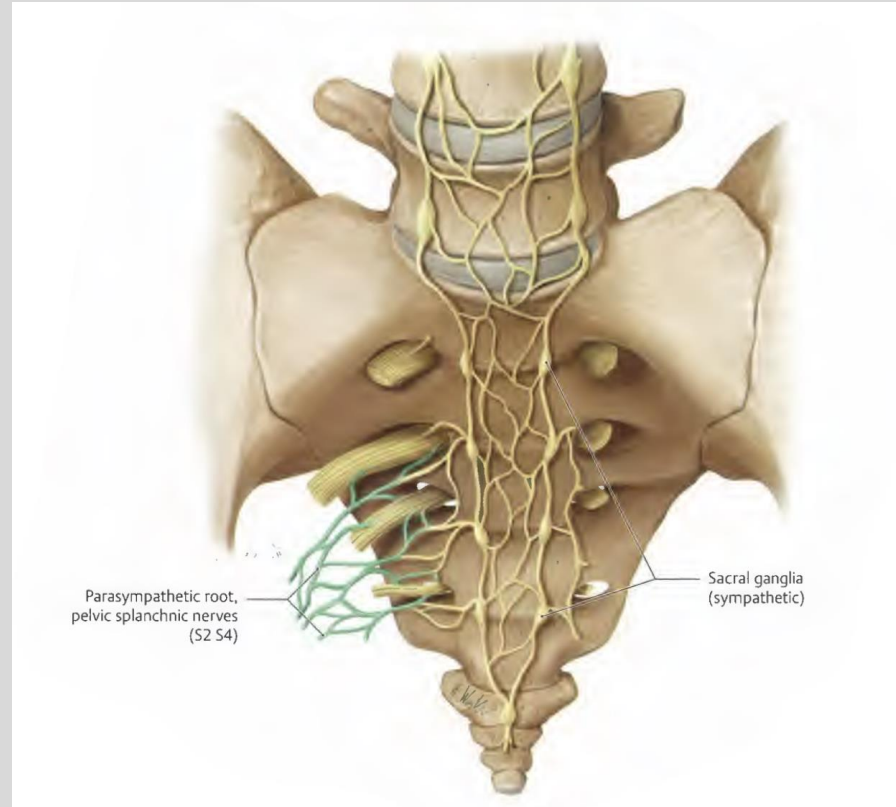
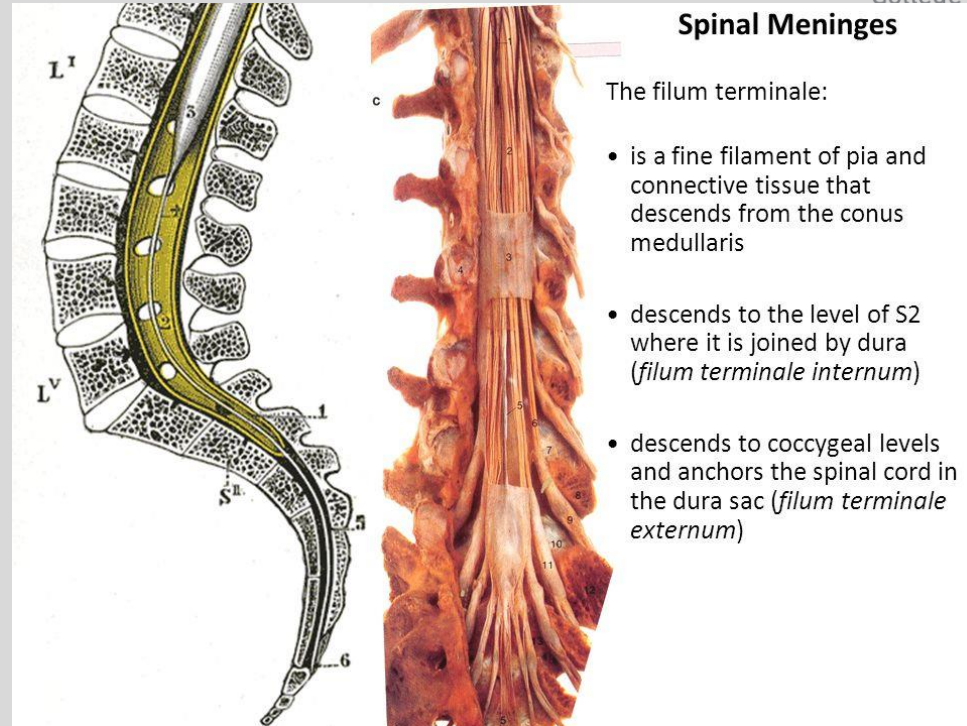


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Caudal Dural Attachments

- S2
- Coccyx1



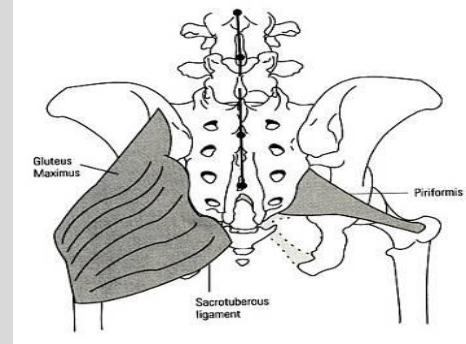
Piriformis muscle

Origin/Insertion: Anterior sacrum to Greater Trochanter of Femur

Action:

- External Rotation
- Extension of thigh
- Abduction of thigh if hip flexed

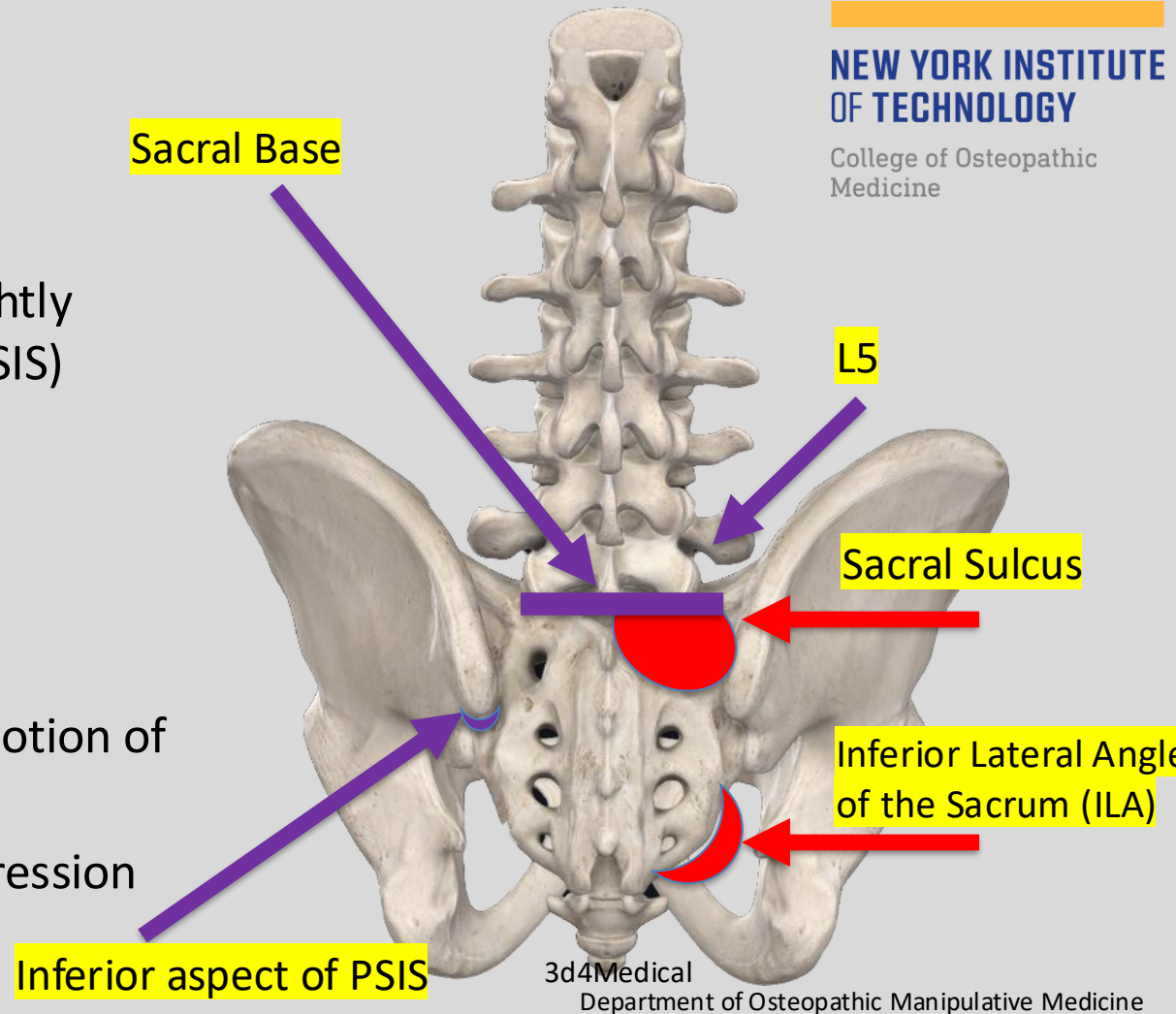
Nerve supply: S1,2



*In 10-11% of population the entire or peroneal portion of the sciatic nerve runs through the belly of piriformis, so if its hypertonic this may result in “**piriformis syndrome**”, pain radiating to posterior thigh, usually not below the knee

- These are the most important landmarks for making a sacral Diagnosis

- Symmetry of:
 - Sacral Sulci (located slightly medial and superior to PSIS)
 - ILAs
- Diagnosis of L5
- Motion at:
 - Lumbosacral junction
 - SI joint with cephalad motion of ILAs
 - SI Joint with ASIS Compression and Seated Flexion test



Axes of Motion of the Sacrum

1 - vertical axis

2 - right oblique axis (gait, sacral torsions and sacral rotation)

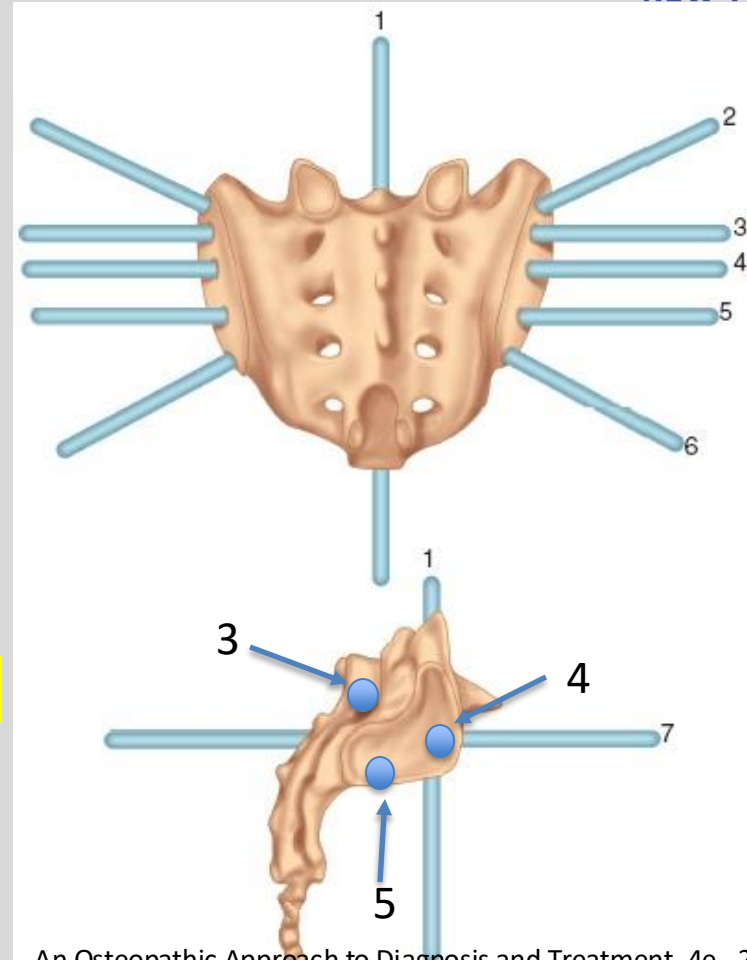
3 - Superior Transverse axis
(respiratory and cranial-sacral motion)

4 - *Middle Transverse*/sacroiliac axis
(Postural Motion)

5 - Inferior Transverse /iliosacral axis
(Gait, Innominate rotation)

6 - left oblique axis (sacral torsions and sacral rotation)

7 - anteroposterior axis



Transverse Sacral Axes

1. **Superior Transverse Axis:** Respiratory and Cranio-sacral motion move around this axis

In Inhalation:

- Sacral base moves posteriorly (& superiorly)
- Thoracic & pelvic diaphragms flatten, diameters increase
- Spinal curves flatten

In Exhalation:

- Sacral base moves anteriorly (& inferiorly)
- Thoracic & pelvic diaphragms dome, diameters decrease
- Spinal curves become exaggerated

2. Middle Transverse Axis

3. Inferior Transverse Axis

Transverse Sacral Axes (Cont')

1. Superior Transverse Axis

- Respiratory and Cranio-sacral Motion

2. Middle Transverse Axis

- Postural Motion (seated or standing)
- Sacral base
 - Anterior (nutation)
 - Posterior (counternutation)
- Forward Bending and Backward bending around this axis

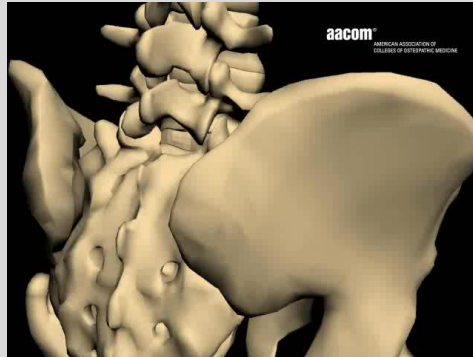
3. Inferior Transverse Axis

Nutation (nodding) vs. Counternutation

Axis: Middle Transverse

Other ways to say these:

- Anterior Nutation vs. Posterior Nutation
- (Sacral) Base Anterior vs. Base posterior
- Bilateral sacral flexion vs. B/L sacral extension



Transverse Sacral Axes (Cont')

1. Superior Transverse Axis

- Respiratory and Cranio-sacral Motion

2. Middle Transverse Axis

- Postural Motion (Sacral nutation/ counternutation)

3. Inferior Transverse Axis - Gait

- Innominate (pelvic) motion on sacrum
- “Dynamic”

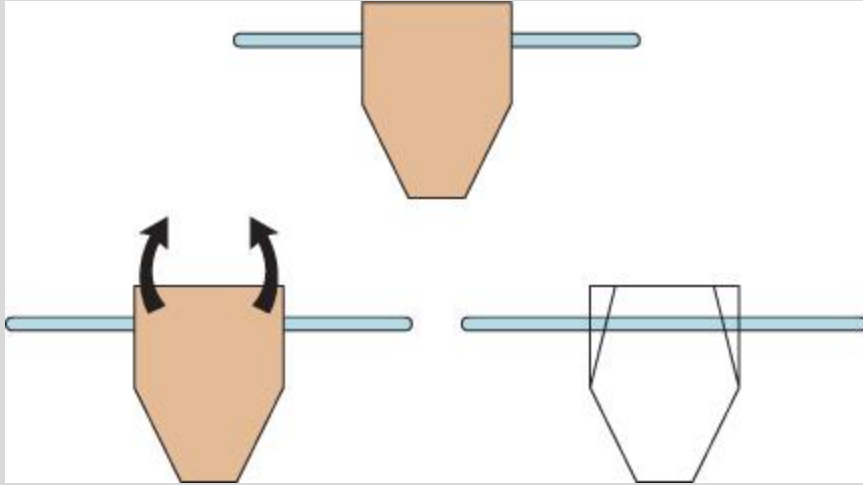
Sacral Biomechanics

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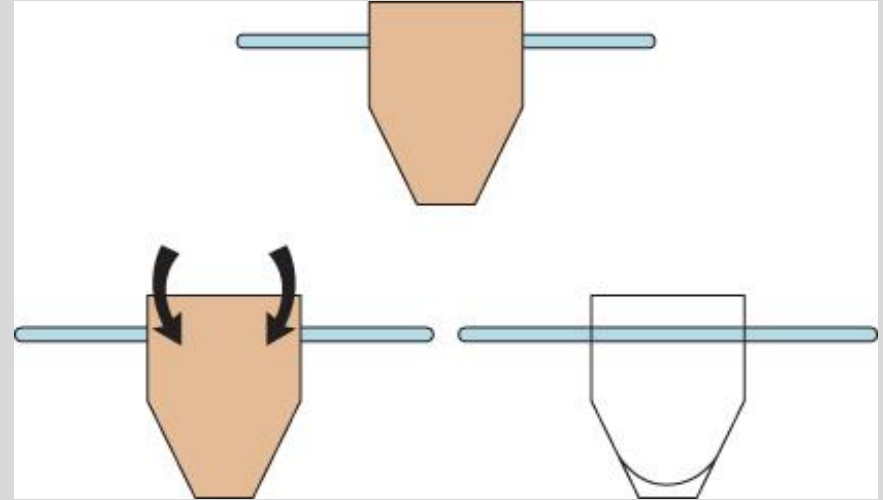
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Bilateral Movement around a Transverse Axis



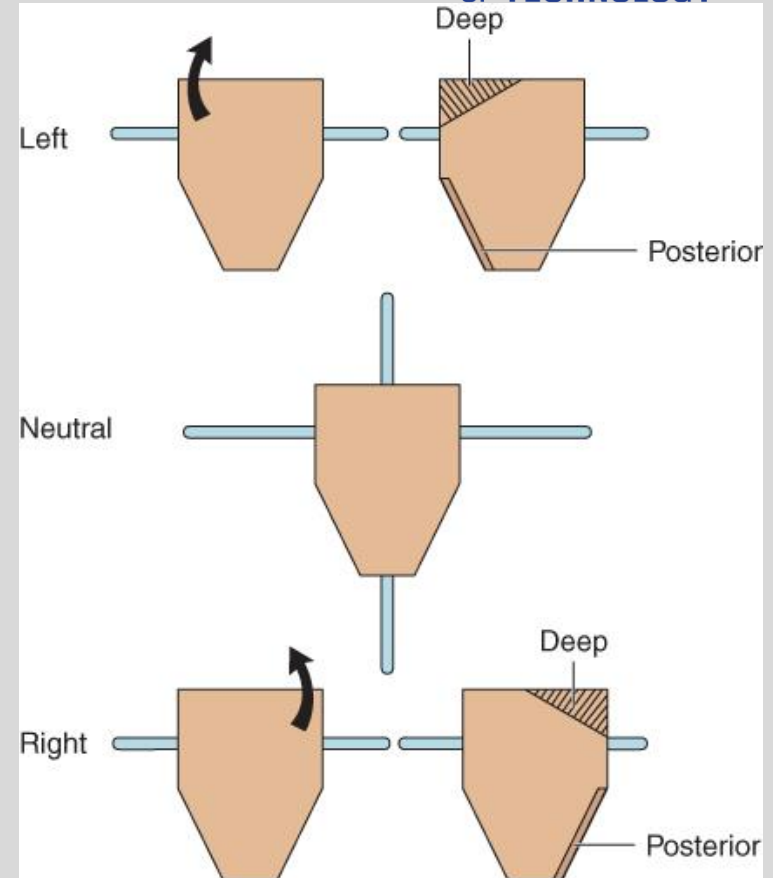
Sacral base is bilaterally flexing over a transverse axis



Sacral base is bilaterally extending over a transverse axis

Unilateral Movement seems to move around a “Transverse Axis?”

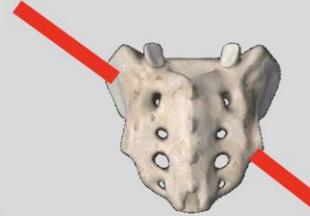
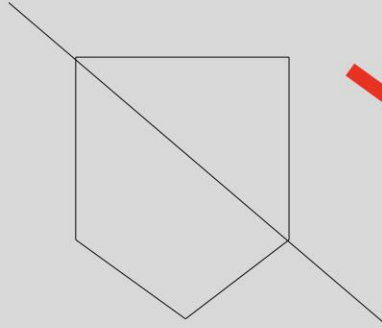
The sacrum can flex or extend unilaterally. The axis for this motion are not well understood. It is not really around a transverse axis. From a palpalatory standpoint, the following occurs. In unilateral flexion, the sacral base on the side of flexion will move anterior and the ILA on the same side will move posterior (and a little inferior). The opposite occurs in unilateral extension



Oblique Sacral Axes

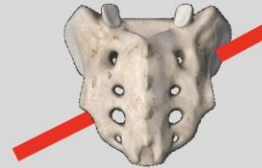
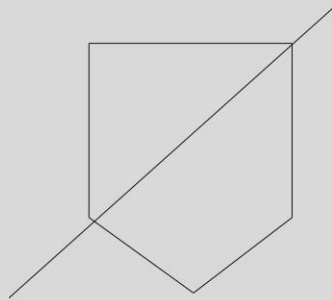
Left Oblique Axis

- Starts from the left superior sacroiliac articulation
- Diagonal to right inferior SI articulation



Right Oblique Axis

- Starts from the right sacral sulcus
- Diagonal to left infero-lateral angle of sacrum

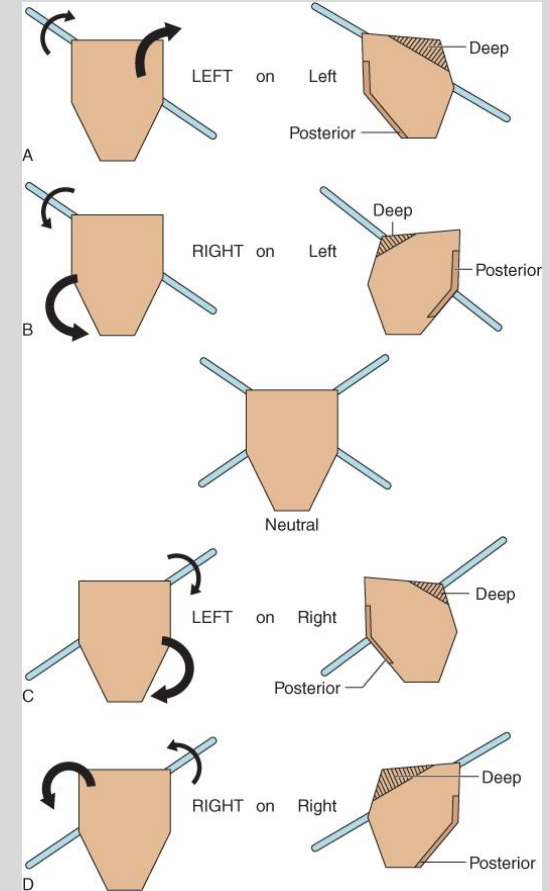
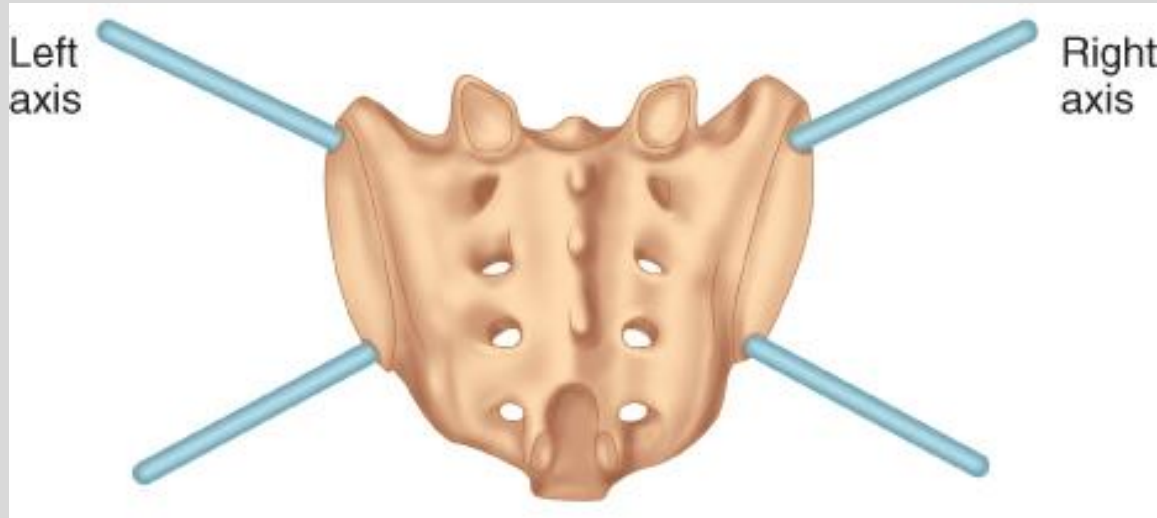


Neutral

Image: 3D4Medical.com

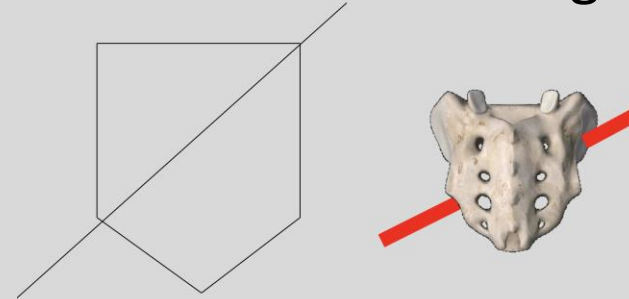
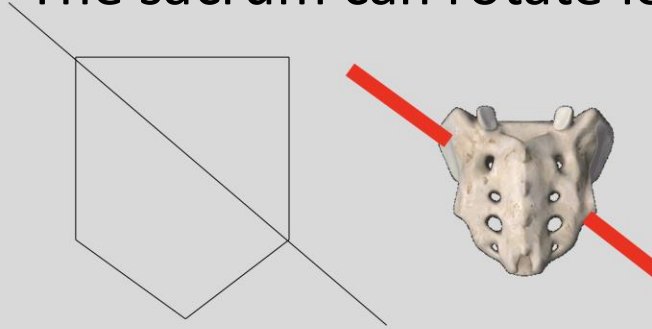
Movement around an oblique axis

Examples are left rotation around a left oblique axis and left rotation around a right oblique axis

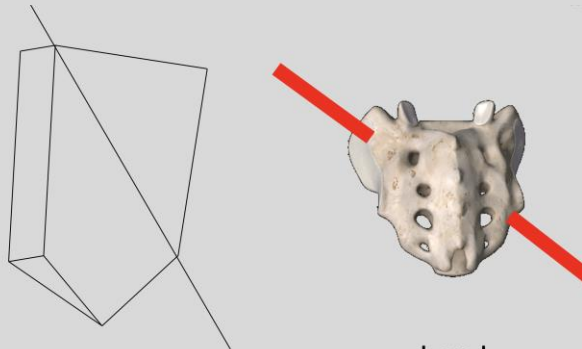


Example: Left Sacral Rotation over oblique axes

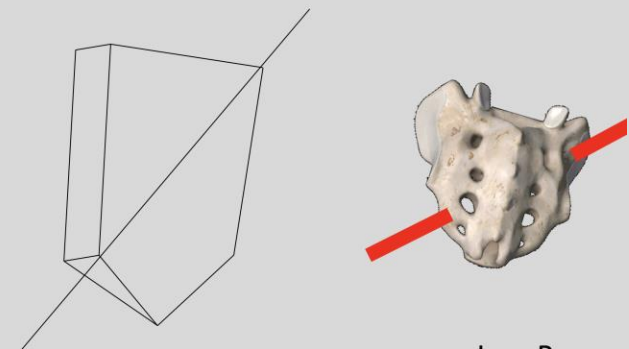
- The sacrum can rotate left on a left axis and left on a right axis



Neutral



L on L



L on R

Osteopathic Diagnosis of the Sacrum

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Prone Static Landmark Examination

+

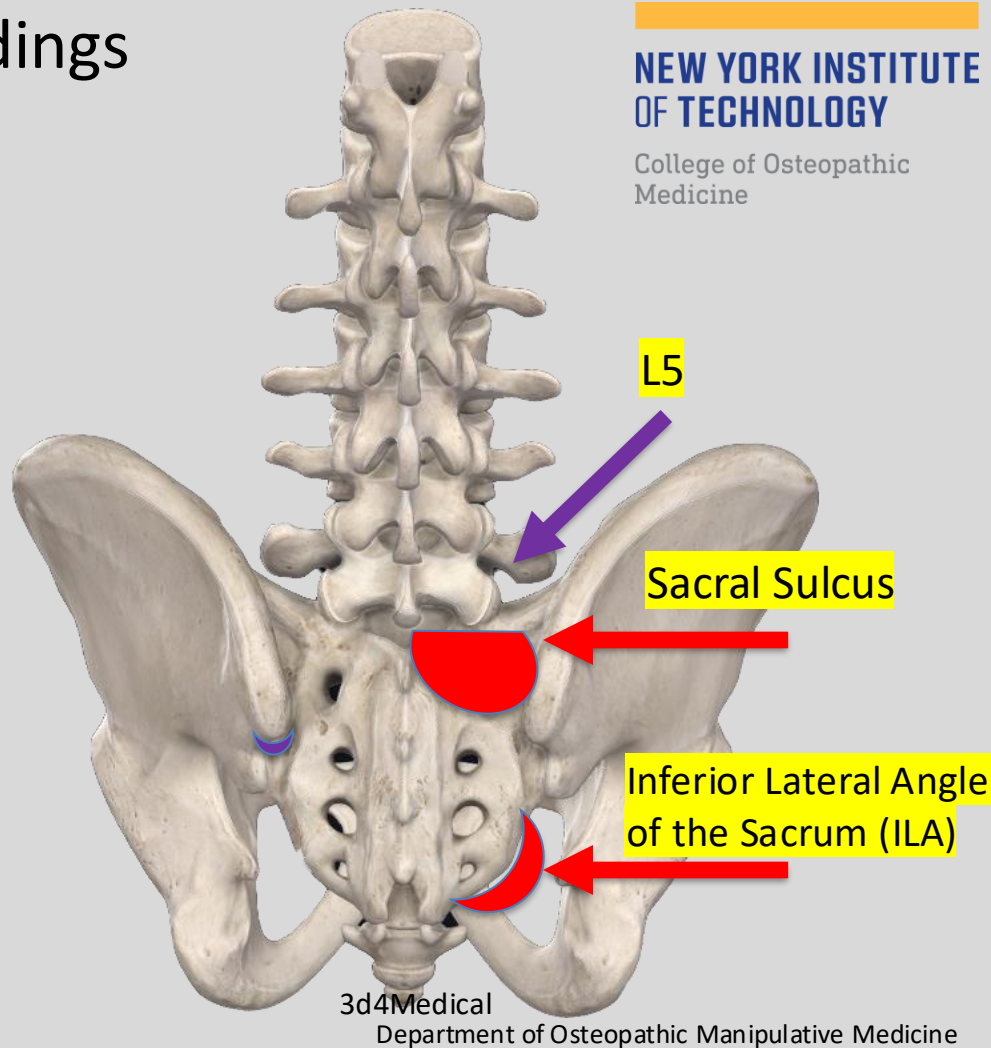
Spring or Sphinx Test

+/-

Seated Flexion Test

Physical Exam: Prone Static findings

1. L5
2. Sacral Sulci
3. Inferior Lateral Angles (ILAs)



- Diagnose L5

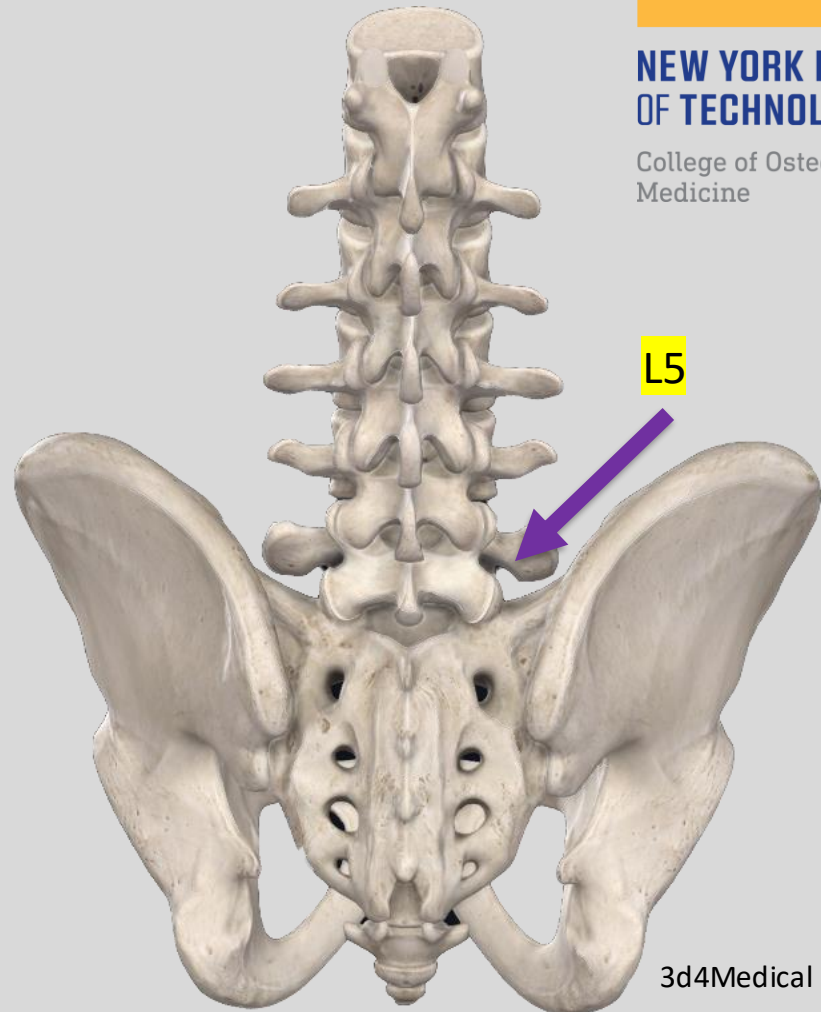
- Direction of L5 rotation influences certain sacral diagnoses:

- Sacral Torsion Somatic Dysfunctions:

- ❖ L5 & Sacrum rotate opposite

- Sacral Rotation Somatic Dysfunctions:

- ❖ L5 & Sacrum rotate same direction



3d4Medical

Check the symmetry of the Sacral Sulci

- Note which side is deeper (sacral base on that side is more anterior) & note which side is more shallow (sacral base on that side is more posterior)
- *Patient position:* prone.
- *Physician position:* varies according to the region examined.
- *Sacral sulci*
 - The physician stands at the side of the table and faces toward the patient's head.
 - The physician places her thumbs on each posterior superior iliac spine (PSIS).
 - The physician hooks her thumbs medially along the PSIS and brings her finger pads into the sacral sulci bilaterally
- The physician evaluates the relative depth of the sulci by two means: By palpating the depth of each sulcus by thumb position.
- By lowering her eyes to the level of the sulci for visual evaluation of depth.



Check symmetry of ILAs

- Note which side is posterior and inferior.
 - Remember if one side is posterior and inferior, the other side is anterior and superior.
1. Patient Prone
 2. Physician's thumbs on inferior aspect of ILAs.
 3. Compare which side is inferior
 4. Then pace thumbs on posterior aspect of ILAs
 5. Compare which side is more posterior



SPECIAL TESTS of the SACRUM

Lumbosacral Spring Test

- **Must be on L-S (lumbo-sacral) JUNCTION**
- **Be mindful of patient**
- **Gentle anterior force**

NORMAL SPRING TEST: If there is spring-ability at the L-S junction this is physiologic= a “*negative*” test:

- Forward moving sacral base (such as seen with forward torsions/rotations, and B/L sacral flexions)

ABNORMAL SPRING TEST = A positive (+) Test = ABSENCE of spring-ability : “steel-like resistance” – non-physiologic. Often result of sacral base being extended or backward

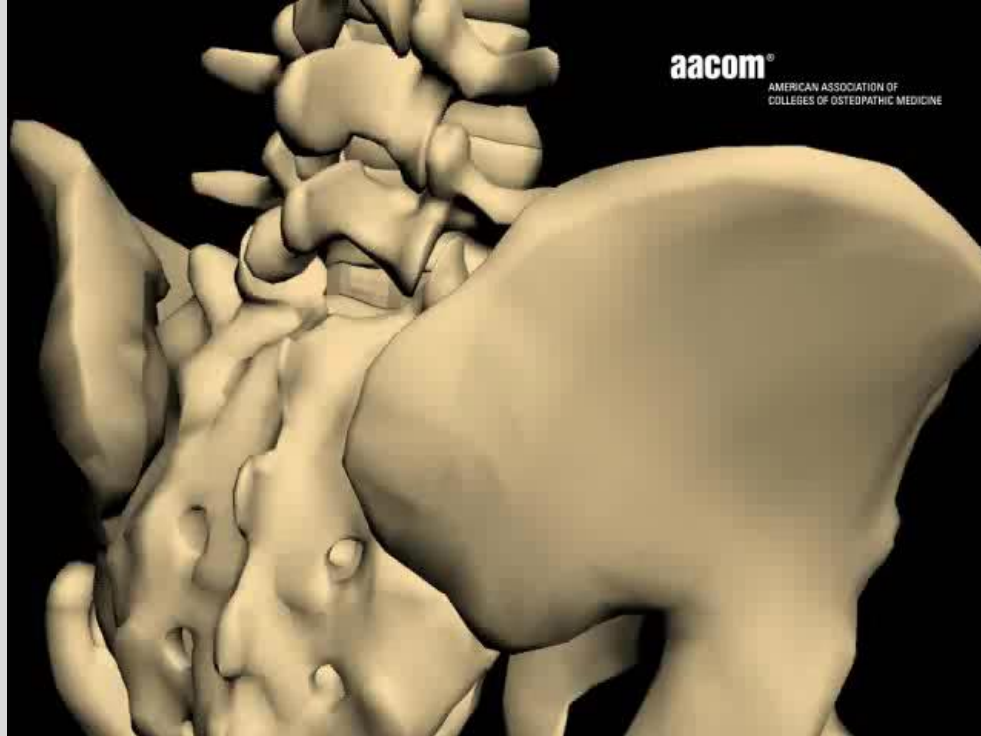


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Sacral Biomechanics

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Sphinx test (backward bending test)

1. Interchangeable with Spring Test

Step 1: Patient prone – make note of static findings (sulci and ILAs)

Step 2: Physician places fingers at sulci and thumbs at ILA's B/L

Step 3: The patient is then instructed to either raise her body onto her elbows or if necessary onto palms to induce extension at the L-S junction. There should be an increased lumbar lordosis and sacral base should go anterior B/L during back extension.

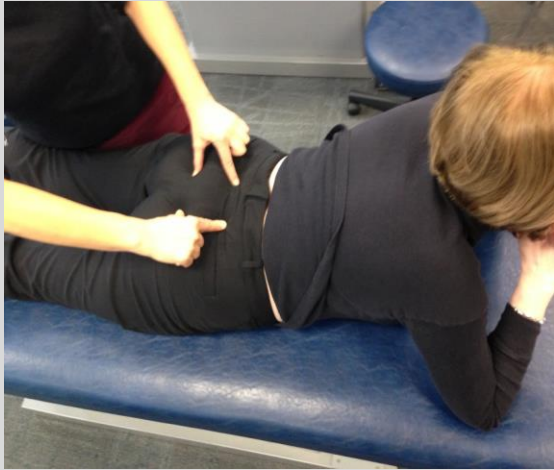
(+) Test = Asymmetrical positioning of ILAs is maintained or worsened. This indicates a NON-physiologic sacral Somatic Dysfunction

(-) Test = ILA's become more symmetric when the patient's back is in extension.

Sphinx test (backward bending test)

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Seated Flexion Test

1. Patient sits with **feet flat on the floor**, knees parted
2. Physician is eye level with patient's PSIS's
3. Physician's thumbs are placed gently on inferior aspect of PSIS bilaterally
4. Patient bends forward from the waist with their arms dropping between their legs (not resting on their thighs)

(+) Test = On the side that your thumb moves **FIRST & FURTHEST**

Indicates Sacral ON Innominate somatic dysfunction (SD)

- Side of positive test is used only in naming of (ipsilateral) unilateral sacral category of SD
- Side opposite of positive test is used for naming the axis for sacral torsions and rotations
- But, the patients don't always follow the rules...will explain in lecture

(-) Test = Either: a) No SD

OR

b) Bilat. SD (B/L flexion or B/L extension SD)



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ASIS compression test

1. Patient is supine
2. Physician places her thenar/ hypothenar eminences along ASIS's B/L
3. Physician alternately introduces gentle but firm downward (A-P) pressure toward one S-I joint at a time from its ipsilateral ASIS contact. Physician allows the ilia to recoil anteriorly against gentle pressure
4. Determine ease of motion and compare bilaterally

(+) Test = Side that offers greater resistance to compression (left or right)

*ASIS Compression test is **NOT specific** for ilio-sacral vs. sacro-iliac dysfunction.



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ILA Motion Testing

(Another Test of Sacroiliac Joint (SIJ) Somatic Dysfunction Laterality)

- Place thenar eminences on the inferior part of ILA's and fingertips at sulci
- Direct force cephalad from ILA into ipsilateral SIJ
- (+) test indicating dysfunction: a decrease in joint play on one or both sides and restriction of sacral motion.
- NOT used for naming sacral dysfunctions but a method of seeing which side may be restricted



Physiologic vs. non-physiologic Categories of Sacral Dysf(x)

- **Physiologic Dysfunction**- somatic dysfunction produces a restriction within the normal range of motion of a joint

Examples:

- Forward Sacral torsions
- Forward Sacral rotations
- Bilateral Sacral Flexions

- **Non-physiologic dysfunction**-restriction of a joint in a position not obtained by usual physiologic motion. Usually traumatic but may occur in a patient with hypermobility

Examples:

- Backward sacral torsions
- Backward sacral rotations
- Unilateral Shears (sacral flexions/extensions)
- Bilateral Sacral Extensions

Non-physiologic dysfunctions should be treated prior to attempting OMT to other diagnoses

Physical exam → diagnosis

1. Seated flexion test (Right or Left or neither?)
2. Sphinx test or Spring test (positive or negative?)
3. L5 Diagnosis (Type 1 or 2? SB? Rotation?)
4. Deep Sacral Sulci (R, L, both, neither?)
5. Posterior/Inferior ILA (R, L, both, neither?)

You will learn how to put these findings into specific Diagnoses in the lab lecture and lab session.

Main Sacral SD Categories

- Bilateral Dysfunctions
- Unilateral Dysfunctions
(a.k.a. “Shears”)
- Sacral Torsions
- (Sacral Rotations)

Main Sacral SD categories w/ the possible outcomes:

1) Bilateral Dysfunctions

- Bilateral flexion (physiologic)
- Bilateral extension (non-physiologic)

2) Unilateral Dysfunctions (both non-physiologic)

- Unilateral Flexion (aka sacral shear)
- Unilateral Extension

3) Sacral Torsions (L5 rotates opposite direction to the sacrum). Sacral torsions are named for “**direction of rotation of sacrum ON direction of the oblique axis**”

- Sacral base is FORWARD (physiologic)
 - L on L (= Left Rotation on a Left Oblique Axis)
 - R on R (= Right Rotation on a right Oblique Axis)
- Sacral base is BACKWARD (non-physiologic)
 - L on R (= Left Rotation on a Right Oblique Axis)
 - R on L (= Right Rotation on a Left Oblique Axis)

4) Sacral Rotations (L5 & sacrum rotate same direction). Similar to sacral torsions, sacral rotations are SD named for “**direction of rotation of sacrum ON direction of the oblique axis**” (same divisions as #3 above)

Pause... We need to mention:

- Sacral Rotations and Torsions have an OBLIQUE AXIS
- UNILATERAL Sacral Dysfunctions – NO AXIS
- BILATERAL Sacral Dysfunctions have a TRANSVERSE (middle) AXIS

Sacral Torsions and Rotations move about an OBLIQUE AXIS

UNILATERAL Sacral Dysfunctions have
NO AXIS

BILATERAL Sacral Dysfunctions move about a middle TRANSVERSE AXIS

NEED TO KNOW

- Sacral Rotations and Torsions have Deep Sulcus and Post-inf ILA on OPPOSITE sides
- UNILATERAL Sacral Dysfunctions have Deep Sulcus and Post-Inf ILA on SAME side
- BILATERAL Sacral Dysfunctions are either flexed (nutated) or extended (counter-nutated)

DIAGNOSING SACRAL TORSIONS BASED ON L5 (You will need to memorized this slide and the next 3 slides. This is a different way of diagnosing sacral torsions which you may see on future exams. In the Lab, we will be going over how to diagnose sacral dysfunctions using the sacral landmarks and and other findings as discussed earlier in this presentation.)

- **L5 will be rotated opposite the direction of the sacrum (if not, L5 must be treated first)**
- **Seated flexion test is (+) opposite the axis of rotation in a torsion.**
- **L5 will be sidebending to the same side as the oblique Axis**

DIAGNOSING SACRAL TORSIONS BASED ON L5 (continued)

- **Forward** torsions are associated with **neutral lumbar** mechanics of **L5** (ie. L5 N SLRR with a L on L FST and L5 N SRRL with a R on R FST)
- **Backward** torsions are associated with **non-neutral lumbar** mechanics of **L5** (i.e L5 ERSR or FRSR with a L on R BST, and L5 ERSR or FRSR with a R on L BST)
- No sacral rotations using this model since the sacrum is rotating opposite L5

What does this mean?

- If you are given that L5 ERS right, this means that L5 has a non-neutral (type 2) dysfunction therefore:
 - The Axis is on the right (the side of the SB)
 - The + seated flexion test is on the left (opposite the axis)
 - Sacrum is rotating left (rotates opposite L5)
 - Diagnosis is Left on Right Sacrum (rotation Left on a Right axis)
 - Left on Right Backward Sacral Torsion (if Rotation and axis are opposite, it is a backward torsion)
 - The same for L5 FRS right

What does this mean? (continued)

- Given that L1-L5 N S left, R right, this means that L5 has a neutral (type 1) dysfunction therefore:
 - The Axis is on the left (the side of the SB)
 - The + seated flexion test is on the right (opposite the axis)
 - Sacrum is rotating left (rotates opposite L5)
 - Diagnosis is Left on Left Sacrum (rotation Left on a Left axis)
 - Left on Left Forward Sacral Torsion (if Rotation and axis are the same direction, it is a Forward torsion)

Session Objectives

1. Be able describe the basic sacral landmarks and their locations
2. Be able to describe how to perform and interpret the different motion tests for sacral diagnosis.
3. Be able to describe the different sacral axes and their significances.
4. Be able to describe physiologic and non-physiologic sacral dysfunctions and what types of sacral dysfunction fall into each category and how they are named
5. Be able to apply the rules of L5 on the sacrum to make a sacral diagnosis if only given an L5 diagnosis.

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

References

DiGiovanna, Amen, Burns. *An Osteopathic Approach to Dx & Tx 4th edition*

Ch 57 “Pelvic and Sacral Anatomic Considerations”

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248813935&bookid=2969>

Ch 60 especially TABLE 60.1 “Evaluation of the Sacrum”

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248814152&bookid=2969>

and as per CPG

Heinking, KP, Giusti, ER, Hensel, KL. *Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research 5th Edition*: Ch 38

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261502266&bookid=3456#261502267>

Sacral Diagnosis

Principles & Practices of Osteopathic Medicine I
OMM Lab #18

Philip Noto, DO

Department of Osteopathic Manipulative Medicine

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

At the conclusion of the session, the students should be able to:

1. Understand the movement of the sacrum on the different sacral axes of rotation and the significance of each axis.
2. Understand L5 rules as they relate to sacral torsions
3. Understand all sacral special tests for diagnosis
4. Be able to put information on axes and special tests together to make a sacral diagnosis

Movement and Axes

- An axis is what movement happens AROUND
- Movement cannot happen ON an axis
- Axes
 - Transverse
 - Obliques

Transverse Axes

- Superior
 - Through S2 attachment of dura
 - Primary Respiratory Motion
 - Gross respiratory motion
- Middle
 - Through S2 vertebral body (Also, center of mass of human body)
 - Sacral motion on pelvis
 - Axis of bilateral sacral flexion or extension dysfunctions
- Inferior
 - Through S3
 - Innominate movement relative to sacrum
 - Axis of innominate dysfunctions

Oblique Axis

- From superior SIJ on one side to inferior SIJ on opposite side
- Sacral torsions and rotations
- Named for movement ON axis
- Reference of movement is the anterior sacral body
- Can be torsioned
 - Left on Left (anterior body MUST be going FORWARD)
 - Right on Left (anterior body MUST be going BACKWARD)
 - Imagine a pair of eyes on anterior sacrum... looking left or looking right)
- Can be rotated
 - Same rotation as side of axis is forward, and opposite is backward (don't need to memorize this!)

Torsions vs. Rotations

- Torsion = Twisting
 - Wringing a towel... twisting means turning ends in opposite directions
 - Sacrum turns in opposite direction of L5
- L5 Rules for sacral torsion:
 - Based on assumption that sacrum is in a torsion...
 - Rotation L5 IS side OPPOSITE of sacral 'rotation' (torsion)- by definition
 - **Side bending IS side of oblique axis**
 - L5 Type I dysfunction = Forward/Type II dysfunction = Backward

Physiologic vs. Nonphysiologic

- Normal motion

- Sacrum moves forward and backward (nutation and counternutation) as well as obliquely during normal everyday movements

- Abnormal motion

Physiologic Dysfunctions:

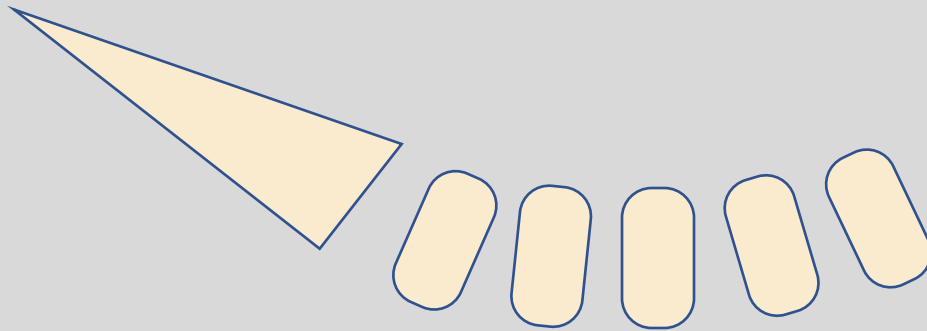
- DYSFUNCTIONAL sacral movement is 'stuck' in a forward direction
- Bilateral sacral flexion dysfunction, forward sacral torsion or rotation

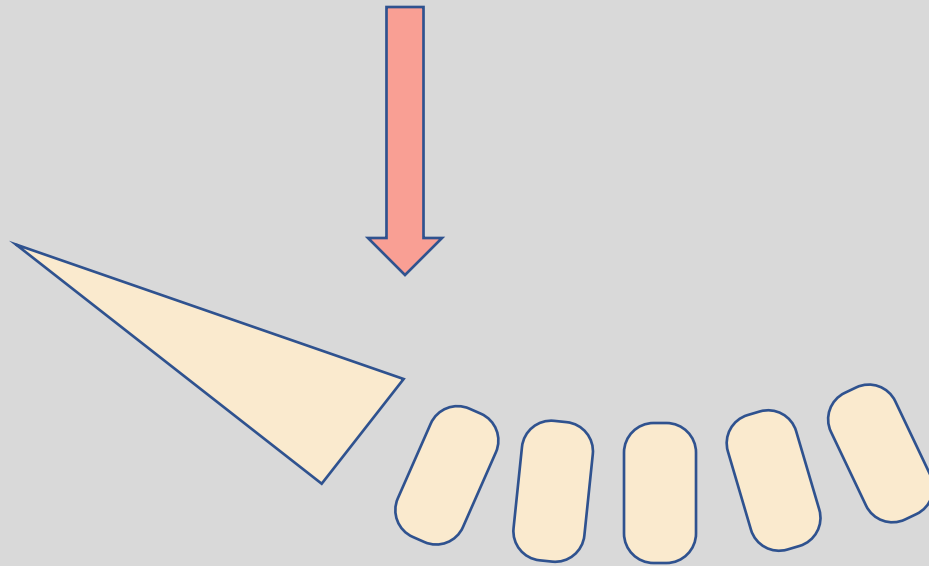
Non-physiologic Dysfunctions:

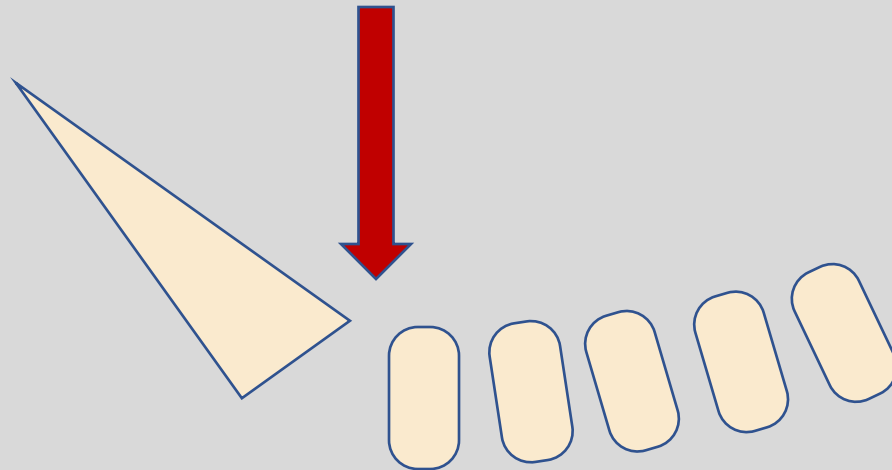
- DYSFUNCTIONAL sacral movement is 'stuck' in a backward direction
- Bilateral sacral extension dysfunction, backward sacral torsion or rotation
- Sacral shears (unilateral)... forward OR backward

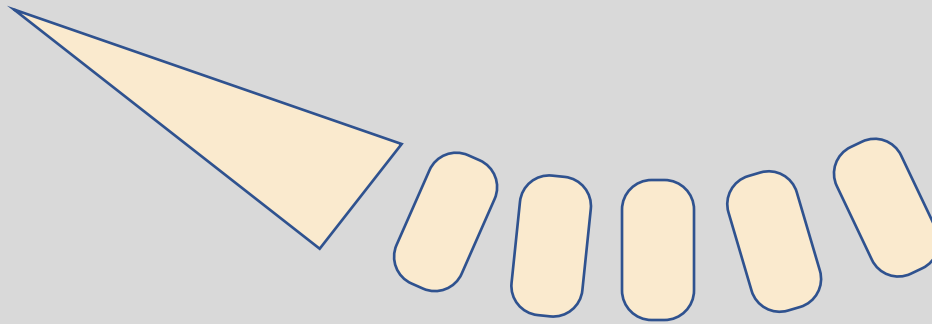
Special Tests

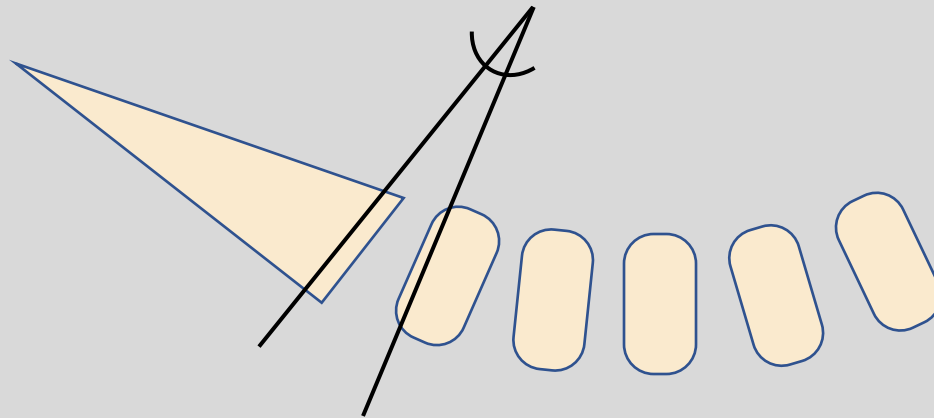
- Spring Test
 - Patient prone
 - Pushing on lumbosacral junction
 - Normal response is a feeling of 'springing' down (forward) and back up
 - If it doesn't spring down/forward (anterior)... it's stuck backward (posterior)
- Sphinx Test
 - Patient prone
 - Palpate at sacral sulci and ILA
 - Patient goes onto elbows or hands → increasing lordosis and lumbosacral angle = sacral flexion
 - If dysfunction is forward, sacral flexion is going into freedom and it evens out
 - If dysfunction is backward, sacral flexion is the barrier and it gets worse

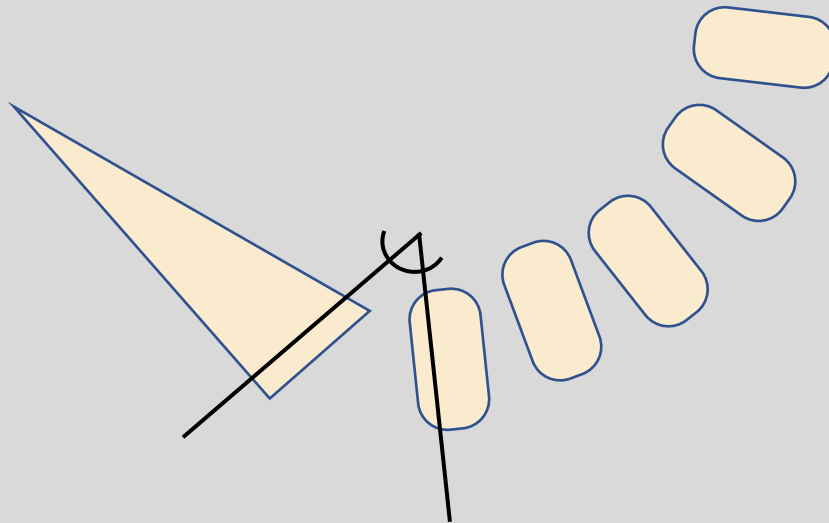






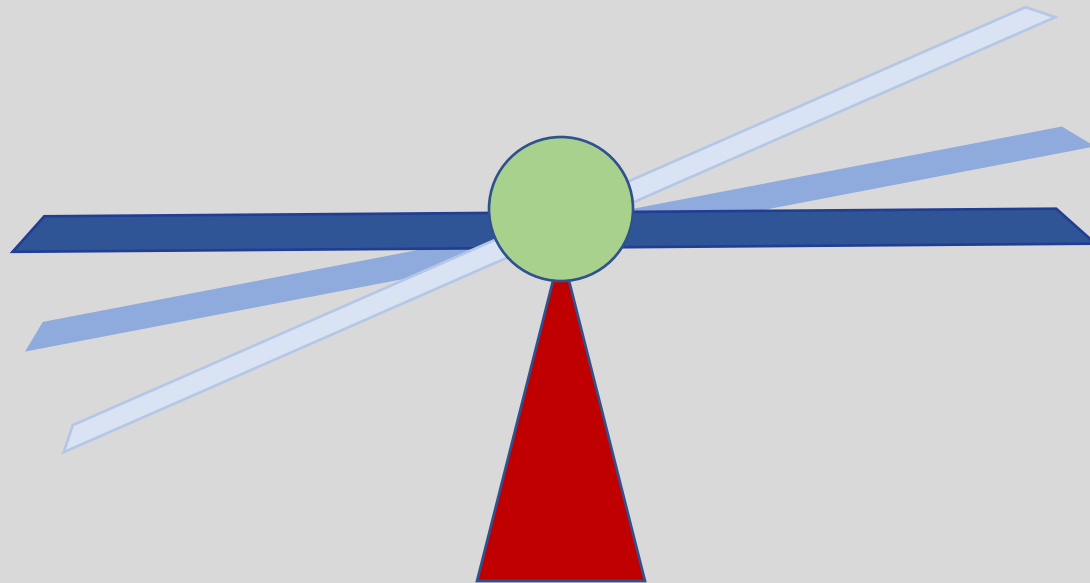


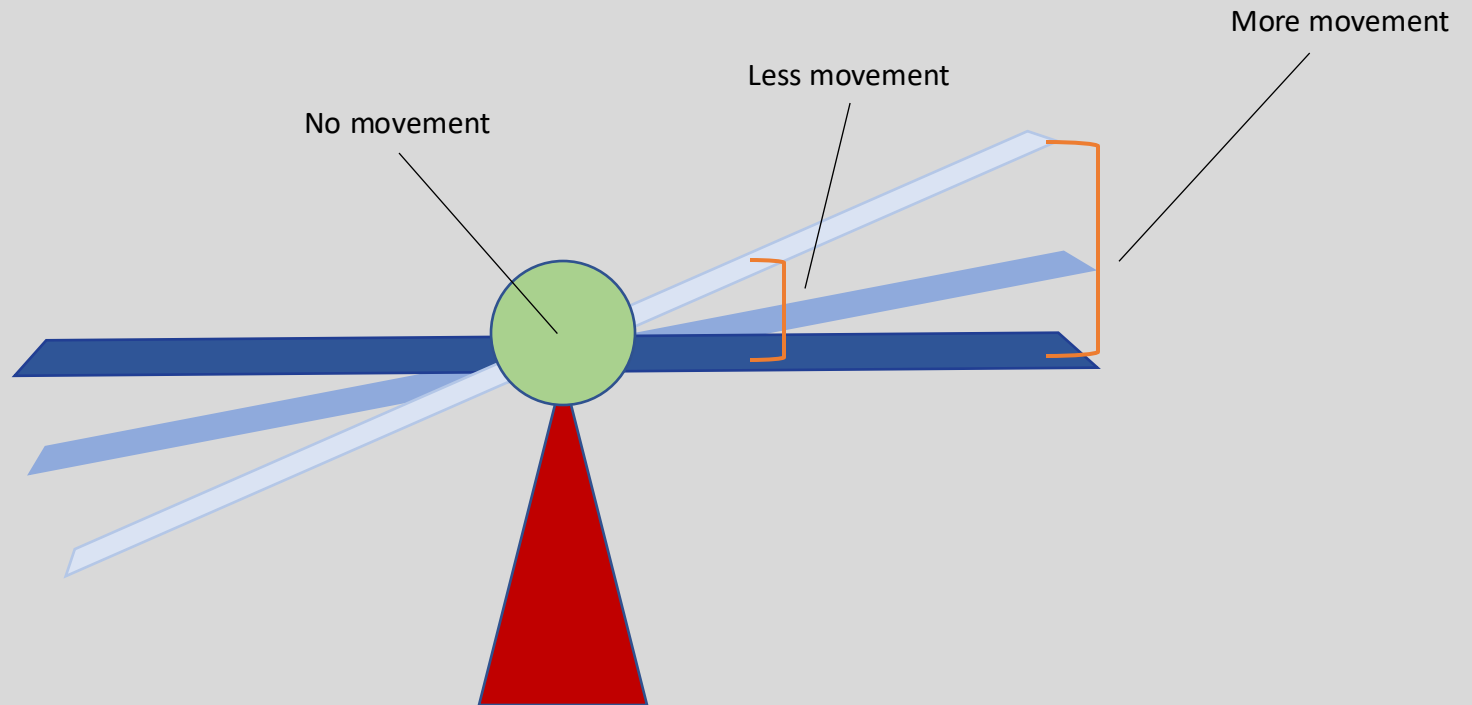


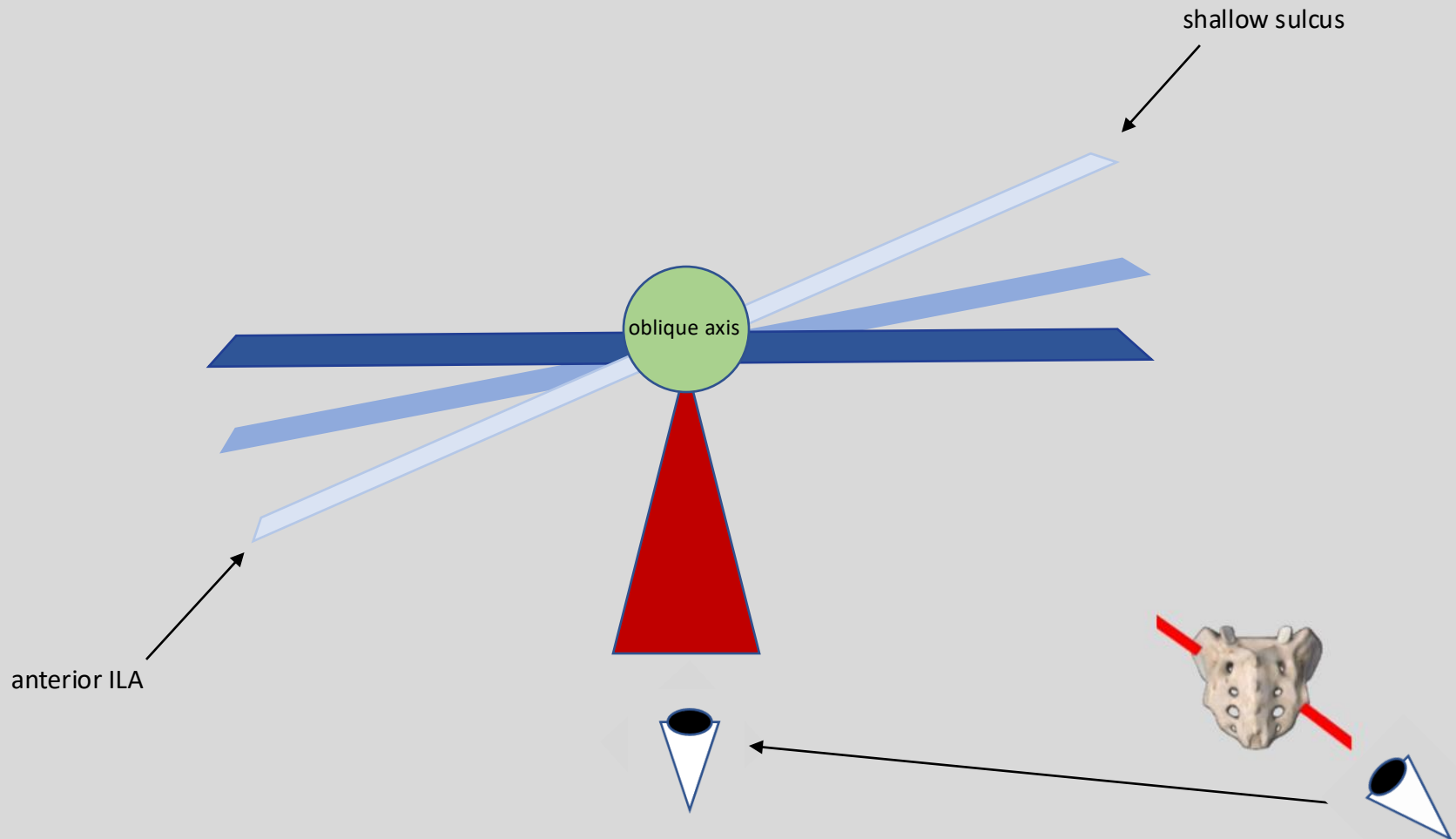


Special Tests

- Seated Flexion test
 - Thumbs on PSIS, on skin
 - Patient seated with feet flat on floor and flexes trunk forward
 - Positive on side of PSIS moving first and farthest
 - This asymmetry indicates some dysfunctional movement on that side
 - Movement happens ON an axis (around it) so axis is on opposite side of positive finding
- Dysfunctional movement...







Application

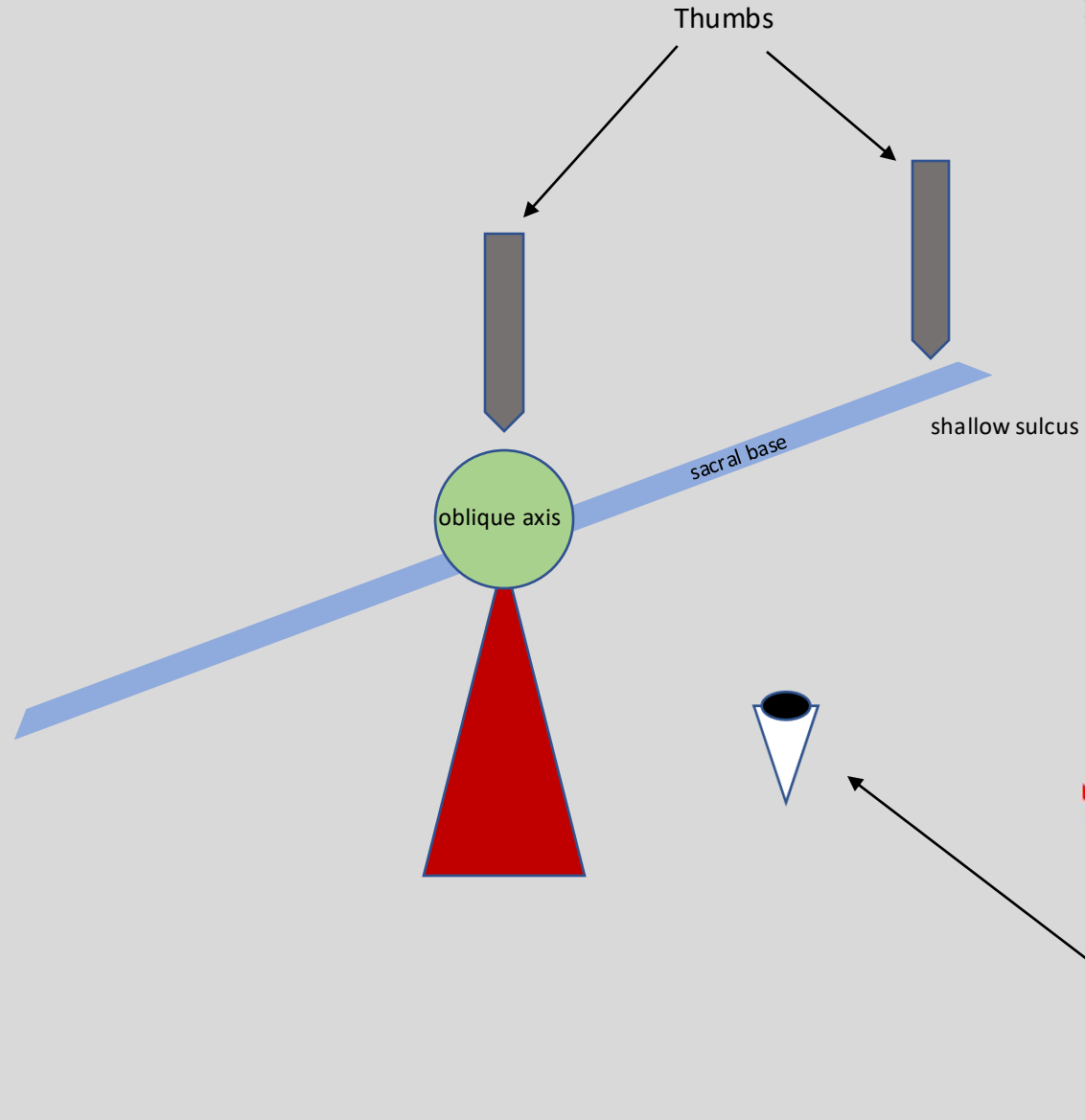
- Seated flexion test: Positive on right / Tells you...
 - Right shear or a some torsion/rotation ON a left axis)
- Spring test: Negative / Tells you...
 - (Forward unilateral flexion or forward torsion/rotation)
- Sacral sulci: Deep on right
- ILA: posterior on left
- L5 NSIRr
- Left (sacrum) ON left (axis) forward sacral torsion

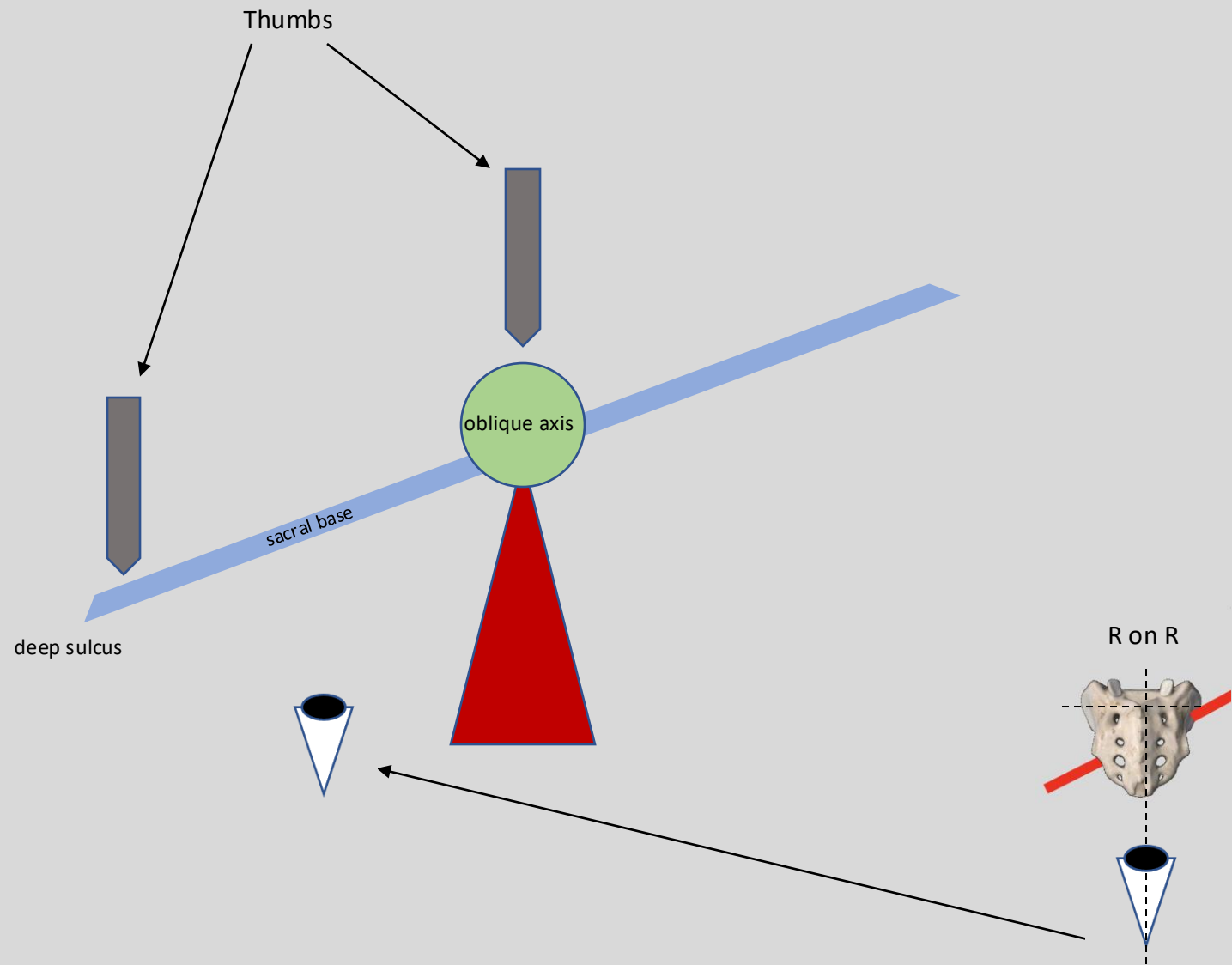
Application

- Seated flexion test: Positive on left / Tells you...
 - Left shear or torsion/rotation ON a right axis)
- Sphinx test: Positive / Tells you...
 - Backward shear or torsion/rotation
- Sacral sulci: Deep on right**
- ILA: Posterior on left**
- Left (sacrum) ON right (axis) backward sacral torsion
- All motion is relative...

Sulci and ILAs

- Motion is relative
 - A 'deep' sulcus is only deep relative to axis bearing sulcus, which feels comparatively shallow
 - A 'posterior' ILA feels posterior only because you're comparing it to the ILA on the other side, which itself feels 'anterior'
 - Your fingers only feel a difference... which finger is right?
- Know whether its forward or backward (Spring or Sphinx)
- Know where axis is (Seated flexion test, L5 position)
- Understand deep vs. shallow relativity
- Apply palpatory findings





Sulci and ILAs

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- Apply palpatory findings

Palpation

- Sulci
 - Thumbs at PSIS
 - Move thumbs medially
 - Probe around this area to appreciate deep vs. shallow
- ILA
 - From sulci 'walk' thumbs down lateral edges of sacrum
 - Feel step off the inferior bony ledge onto soft tissue
 - Palpate pressing anteriorly on that ledge before stepping off
 - Palpate superiorly against that ledge from it's inferior aspect

List of diagnoses

- R on R (right on right forward sacral torsion or rotation)
- L on L (left on left forward sacral torsion or rotation)
- R on L (right on left backward sacral torsion or rotation)
- L on R (left on right backward sacral torsion or rotation)
- LUSF/RUSF (left or right unilateral sacral flexion)
- LUSE/RUSE (left or right unilateral sacral extension)
- Bilateral sacral flexion
- Bilateral sacral extension

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- Seffinger MA. *Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research*. 4th ed. Philadelphia, PA: Wolters Kluwer; 2018.

Lab Feedback Form

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